



Spinel

△ **Octagonal** mixed-cut ruby spinel

Gemstone spinel is a magnesium aluminium oxide, although the name is also given to a group of metal oxides, all of which have the same crystal structure. It is most familiar as a blue, purple, red, or pink gem, but it can occur in other colours; its blood-red variety is sometimes called “ruby spinel”. Another variety, star spinels, are so named as they display stars created by natural light reflection within the gem. Most spinel is recovered from stream gravels; the earliest gem dates from 100 BCE and was discovered in a Buddhist tomb.

Specification

Chemical name Magnesium aluminium oxide | **Formula** $MgAl_2O_4$
Colours Red, yellow, orange, red, blue, green, black, colourless
Structure Cubic | **Hardness** 8 | **SG** 3.6 | **RI** 1.71–1.73 | **Lustre** Vitreous | **Streak** White | **Locations** Myanmar, Sri Lanka, Vietnam, Madagascar, Afghanistan, Tajikistan, Pakistan, Australia, Tanzania

Rough



Spinel group | These gem-quality spinels shows the mineral's colour range. Some specimens are water rounded, while others have complete octahedral crystal forms.

Single crystals



Aggregate | This uncut stone is an aggregate of a number of small, gemmy, red spinel crystals that have naturally bonded together.

Octagonal crystals



Magnetite | The distinctive, dark-coloured iron oxide magnetite is one of the spinel group of minerals, seen here as a cluster of black octahedra.

The Black Prince's “ruby”

A case of mistaken identity

The superb spinel known as the Black Prince's Ruby was supposedly given to Edward, the Black Prince, the son of English King Edward III, by Peter the Cruel, king of Castille, after their joint victory at the Battle of Najera in 1367. Another English king, Henry V, wore it and nearly lost it at the Battle of Agincourt in 1415, and it was later set in the British Imperial State Crown. It was thought to be a ruby until the 19th century.



The “ruby” Shown here in its diamond-encrusted mount, the spinel is set with a small natural ruby near the top.